

## Review

## Vision-based excavator pose estimation for automatic control

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## ABSTRACT

Hydraulic excavators, widely employed in harsh environments, have garnered increased attention in recent years from manufacturers and researchers for automatic operation, particularly through vision-based manipulator pose measurement. This paper utilized a mixed review method to explore the vision-based measurements as well as other applications. A quantitative analysis of computer vision (CV) applications on excavators was performed through a literature search, leading to the classification of these applications into three categories based on keywords. Subsequently, a comprehensive literature review identified four key requirements and challenges for vision-based measurement in the context of automatic control, followed by a discussion on the future prospects of marker-based and deep learning-based manipulator pose measurement for automatic control. This paper offers an in-depth examination of CV applications in excavators, emphasizing the pertinent challenges and forthcoming trends in vision-based manipulator pose measurement for automatic control.

## 1. Introduction

Hydraulic excavators are widely utilized in many fields, including mining, forestry, and construction, due to their efficient, versatile, and robust performance [1]. In addition to digging soils for creating trenches or holes, these machines can also be employed for leveling operations, heavy lifting, and transportation [2,3]. However, construction sites present numerous harsh and hazardous working environments that may lead to severe injuries to workers and property loss in case of accidents [4]. As a result, the interest for automatic excavators has increased observably among manufacturers and researchers of earth-moving machines over the past two decades [5]. The utilization of automatic excavators in such hazardous environments offers improvements in productivity and safety.

In the field of automatic excavators, precise control for the operation trajectory is a fundamental technology. The operation trajectory refers to time-dependent positions and orientations of the end effector relative to its base, which are determined by the real-time status of all links in the manipulator (manipulator pose). Therefore, real-time measurement of manipulator pose is a prerequisite for automatic control of operation trajectory. The traditional approaches are contact-sensor-based methods, which employ sensors installed on the manipulators to measure their pose. These contact sensors include the inertial measurement unit (IMU) [6,7], cylinder displacement sensor [8], and other sensor

types [9,10]. However, there are some limitations to contact sensors. For instance, noise and vibration are evident in the harsh working environment, which can damage these contact sensors. Moreover, the installation and maintenance are costly, time-consuming, and laborious. Due to the advantages of low cost, convenient installation, and non-contact measurement, vision sensors are increasingly being employed in engineering equipment [11]. Therefore, computer vision (CV) based measurement of excavator manipulator pose is a potential method.

In recent years, the applications of CV in industrial fields are increasing to improve the level of automation. CV is an interdisciplinary technology that deals with how computers analyze digital images and videos, aiming to achieve a high-level understanding of the objects presented in the images [12]. Over the past decade, CV for excavator automation has been attracting increasing attention from researchers. The purpose of CV applications is to monitor various aspects of working environments, including the detection and tracking of the objects to be excavated [13], the recognition and positioning of trucks [14] and other excavators [15], and the detection of workers [16] and other devices [17] on construction sites. It can also be employed to perceive the state of excavators, such as excavator action recognition [18–20], parts detection [21] and manipulator pose estimation [22,23]. Among these applications, manipulator pose estimation is crucial for the automatic control of operation trajectory. However, most existing vision-based measurement methods are primarily used for safety monitoring, which

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cannot be directly used as a feedback step for automatic control. To the best of the authors' knowledge, there is no relevant systematic literature review available currently. Therefore, a comprehensive review is necessary to address these knowledge gaps and provide insights into the current research status of vision-based manipulator pose measurement, as well as the challenges associated with feedback control. This paper aims to propose research ideas for the development of vision-based manipulator pose measurement methods, thereby enabling vision-based automatic operations. The research encompasses the following aspects. (1) A comprehensive exploration of vision-based methods employed on excavators is presented, covering not only manipulator pose estimation but also other CV applications. This paper provides both quantitative and qualitative analysis of these methods. (2) The challenges and future developments of vision-based manipulator pose measurement methods are discussed from the perspective of automatic control of operation trajectory.

This paper is organized as follows. Section 2 introduces the mixed review method employed in this paper and presents a detailed account of the literature search conducted to identify CV applications on excavators. It also provides a quantitative analysis of the results obtained. In Section 3, a comprehensive review of the identified literature is presented. Section 4 focuses on discussing the requirements and challenges for vision-based pose measurement methods from the perspective of automatic control of manipulator operation trajectory. The future developments in this field are also discussed. The structure of this paper is illustrated in Fig. 1.

## 2. Mixed review method

Many researchers [4,24–26] have employed a mixed review method consisting of quantitative and qualitative analysis (i.e., bibliographic

analysis and systematic review). The objective of employing the mixed method is to utilize the strengths and minimize the weaknesses of both qualitative and quantitative methods [27]. Bibliometric analysis is a statistical method used to visualize the structural and dynamic characteristics of scientific research [28]. A systematic review is conducted to offer a comprehensive overview of existing research for the purpose of identifying gaps in the body of knowledge and anticipating future research directions. Prior to using the mixed review method, the purpose and conceptual boundaries of this paper are analyzed. The complete research method used in this paper is presented in Fig. 1 and consists of five consecutive steps as follows.

step1: Identify the purpose of the paper (Section 2.1).

step2: Identify the conceptual boundaries of the paper (Section 2.2).

step3: Data collection (Section 2.3).

step4: Bibliographic analysis (Section 2.4).

step5: Systematic review (Section 3 and Section 4).

### 2.1. Identify purpose of the paper

CV has introduced new working modes in excavator operations, while also presented new challenges and issues for its specific applications. Currently, CV technology has been employed for excavator detection, operation action analysis, manipulator pose estimation, and monitoring operational environments. These applications mainly focus on safety monitoring and production management for semi-automated or remote operation. However, there are only a few instances where CV is employed for operation trajectory automatic control, despite its potential in this area. This paper aims to provide valuable research ideas for the development of vision-based measurement methods for excavator manipulator pose, ultimately enabling the realization of vision-based automatic operations.

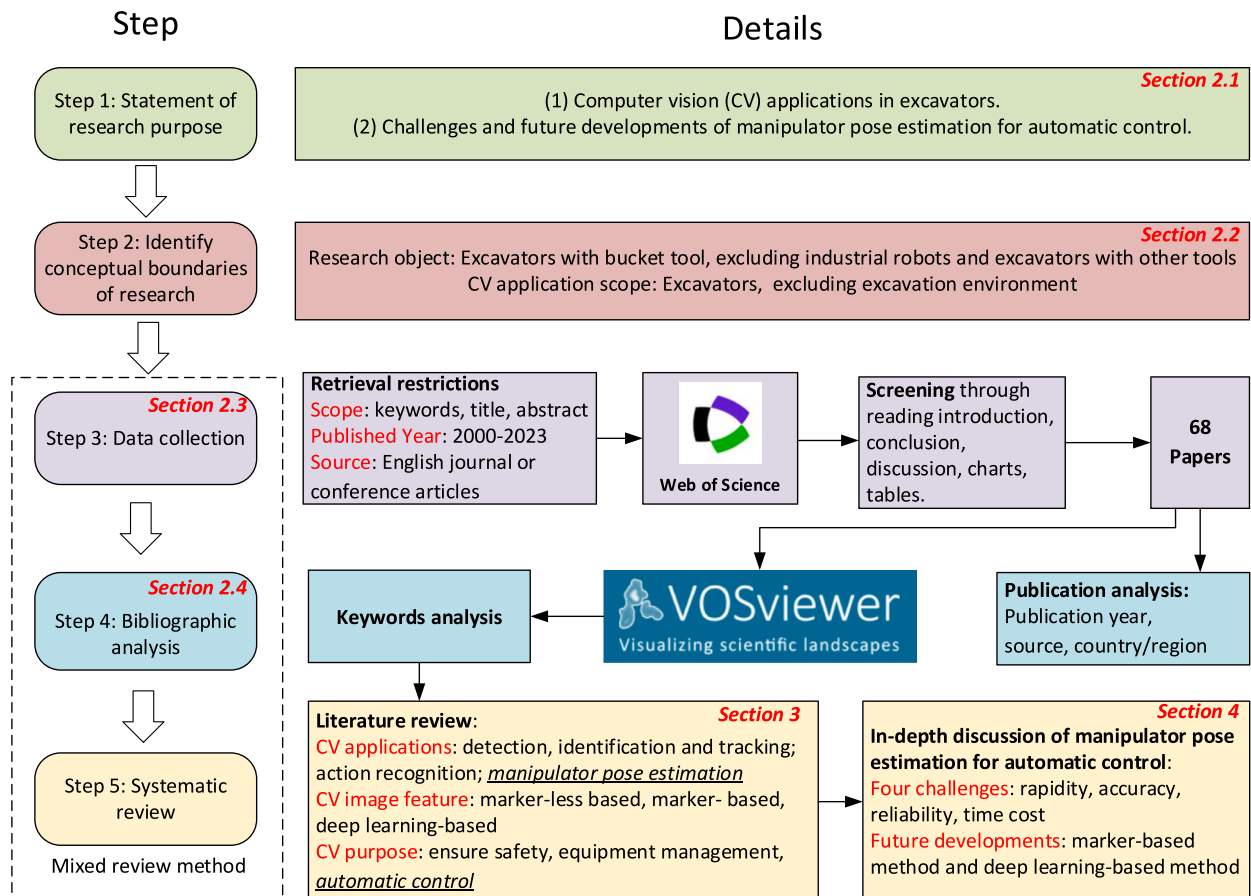


Fig. 1. Chapter structure and research structure.

## 2.2. Identify conceptual boundaries of the paper

(1) It is important to note that while heavy-duty hydraulic excavator manipulators share similarities with conventional industrial robot arms [29], there are differences in the application of CV between them, which will be explained in Section 4.1. Therefore, this paper focuses on hydraulic excavators, excluding industrial robots. (2) Excavators can be equipped with various working tools, including buckets, hydraulic breaking hammers [30], hydraulic shears, hydraulic rippers, breaking hammers, and other tools. Among these tools, the bucket stands out as the most commonly used and extensively studied. This observation is supported by the research results. For instance, a search using the keyword “excavator” in the Web of Science core dataset retrieves 3213 English-language journal articles and conference papers. The majority of these publications primarily focus on the use of the buckets as the main working tools. A search for “hydraulic hammer” OR “hydraulic breaker” OR “hydraulic breaking hammer” (including quotation marks) yields only 90 papers, while a search for “hydraulic shear” results in 165 papers. However, there is scarce literature related to CV among these sources. Therefore, this paper primarily concentrates on excavators that employ buckets as their primary working tool. Unless explicitly stated otherwise throughout the subsequent chapters, the term “excavator” in this paper refers specifically to excavators equipped with buckets. (3) This paper emphasizes the applications of CV in monitoring excavators themselves, such as excavator detection and estimation of their operating pose. Environmental detection, such as identifying rocks, soil, or other equipment at the construction sites, is not within the scope of this paper.

## 2.3. Data collection

Firstly, a commonly used academic database covering multiple comprehensive fields was selected, and search keywords were determined to delimit the research area. Subsequently, a preliminary screening of the literature was conducted based on the keywords, titles, and abstracts. After that, a secondary screening was performed, involving a thorough review of the introductions, conclusions, discussions, charts, tables, and full-text contents to select papers for further study.

### 2.3.1. Literature database

Considering its comprehensive coverage of a wide range of journal publications and knowledge fields, Web of Science [31] was chosen as the database for literature search in this paper, as opposed to other databases.

### 2.3.2. Data collection

To ensure high-quality and reliable research data, careful selection of references is necessary. Therefore, the preliminary data collection in this paper was constrained by the following criteria. (1) Theme constraint: The main keywords were “excavator OR excavators” and “machine vision OR computer vision OR vision OR visual OR image process” to focus on excavator-related research while exclude other aspects. These two sets of keywords were connected by the logical operator “AND”. (2) Publication year constraint: The analysis included papers published from 2000 to the present (2023) to explore recent advancements in the field. (3) Language constraint: Only English literature was analyzed to facilitate reading and analysis. (4) Quality constraint: Peer-reviewed articles from journals and international conferences were selected to ensure a quantitative and objective review. The search properties and their corresponding values are presented in Table 1.

### 2.3.3. Screening

According to the search criteria outlined in Table 1, a total of 178 papers related to CV in excavators were obtained from Web of Science Core collection. It was found that many of these papers only briefly mentioned the topic of “excavator computer vision”. Therefore, a

**Table 1**

Search properties and their values about literature search.

| Properties     | Values                                                                                                       |
|----------------|--------------------------------------------------------------------------------------------------------------|
| Database       | Web of science                                                                                               |
| Keywords       | (excavator OR excavators) AND ((computer vision) OR (machine vision) OR vision OR (image process) OR visual) |
| Search scope   | Article title, abstract, keywords                                                                            |
| Published year | From 2000 – present (2023)                                                                                   |
| Source         | Journal articles / conference articles                                                                       |
| Language       | English                                                                                                      |

detailed review of the titles, keywords, and abstracts of all 178 papers was conducted. Papers unrelated to excavator CV applications, such as excavator operating environment detection, teleoperation of excavators, and image segmentation of the target to be excavated were excluded. As a result, 96 papers were retained. Subsequently, a thorough screening of the introductions, conclusions, discussions, charts, tables, and other key information in these 96 papers was performed. Finally, 68 papers were selected for bibliometric and systematic review in the subsequent sections.

## 2.4. Quantitative analysis

This section presents a quantitative analysis of 68 papers. Firstly, we conducted a statistical analysis of the publication years to examine the trends and factors driving the development of CV in the field of excavators. Next, we discussed the sources and affiliations of these papers. Finally, we summarized the main areas based on the occurrence frequency of keywords in order to gain insights into the current research status.

### 2.4.1. Year of publications

The first paper on CV applied to excavators was published in *Journal of Computing in Civil Engineering* in 2007 [32]. In the reference [32], the image captured from a remote camera was processed by a computer using the Hue, Saturation, Value (HSV) color space to distinguish the excavator from the background. The excavator was detected at intervals of 10 s to determine its operational status and estimate the idle time. As shown in Fig. 2, prior to 2012, there was limited research on the application of CV in excavators, possibly due to the early stage of development of image processing algorithms, cameras, high-speed computers, and other technologies. From 2013 onwards, researchers gradually started applying CV to construction equipment. Moreover, there has been further advancement closely associated with the advancements in artificial intelligence [33], image processing technology [34], and deep learning technology [35] since 2018. It is worth noting that the literature analysis for this paper was conducted in January 2023, resulting in a smaller number of papers from 2023 compared to the entirety of 2022. In conclusion, the research interest trend is evident, indicating that CV-based excavator automation will continue to be a prominent research area in the coming years.

### 2.4.2. Sources

Most of the papers were published in the fields of construction and civil engineering. Out of these 68 papers, 22 were from *Automation in Construction*, and 10 were from *Journal of Computing in Civil Engineering*, accounting for 32.35% and 14.71% of the total papers, respectively. Additionally, there were 18 papers presented at 18 different international conferences, and the remaining 18 papers were published in 15 different journals, including *Multimedia Tools and Applications*, *Advances in Civil Engineering*, *IEEE Access*, *Sensors*, and *Advanced Engineering Informatics*. As depicted in Fig. 3, *Automation in Construction* and *Journal of Computing in Civil Engineering* are the two most popular journals in this field, collectively comprising nearly half (47.06%). Moreover, *Automation in Construction* has received more than 770 citations, while *Journal*

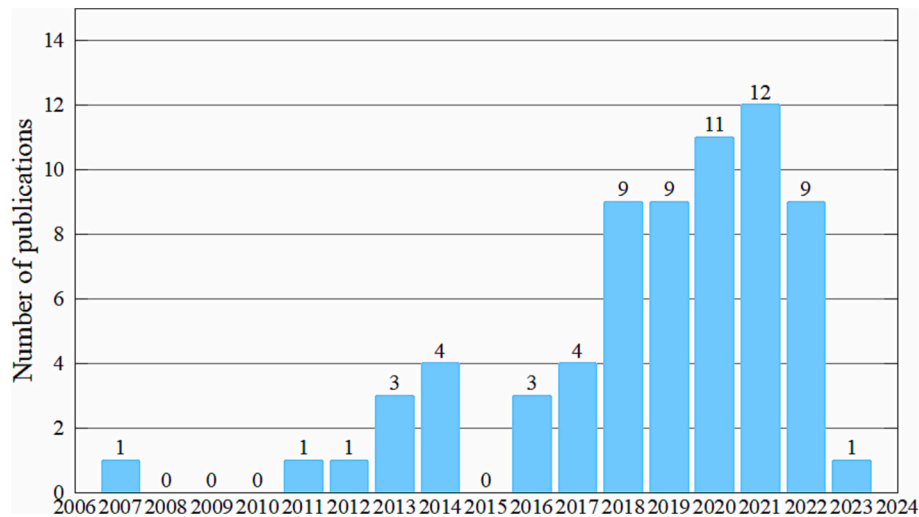


Fig. 2. Publication years of literature.

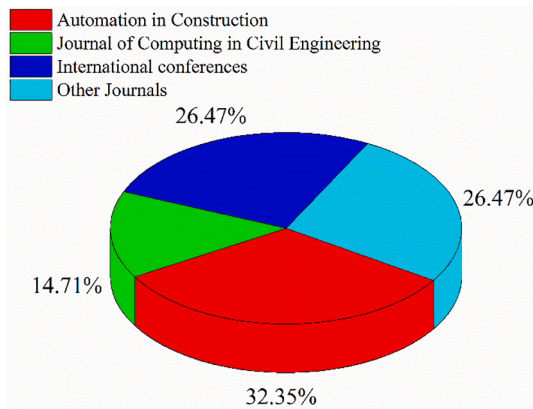


Fig. 3. Sources of literature.

of *Computing in Civil Engineering* has amassed over 370 citations. This highlights the influential role of these two journals in the application of CV in excavators.

#### 2.4.3. Publications distributed by country/region

The selected papers were analyzed based on the affiliation of authors. As depicted in Fig. 4, Chinese authors were accounted for 26.47% (18 papers) of the total 68 papers, followed by authors from the United States (23.53%), South Korea (22.06%), and Canada (13.24%). Authors

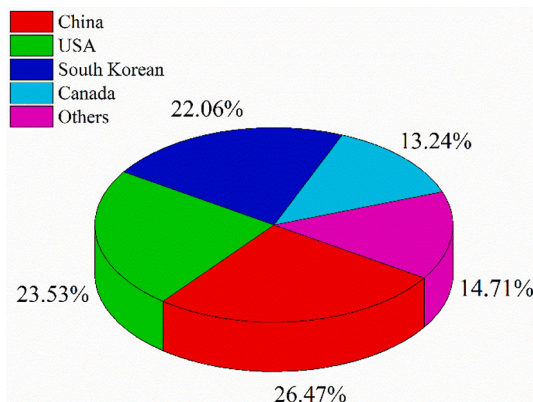


Fig. 4. Publications distributed by country/region.

from nine other countries and regions accounted for the remaining 14.71%.

#### 2.4.4. Keywords

Keywords play a crucial role in understanding the research content and trends of relevant papers. In this section, we employed VOSviewer [36] to analyze the occurrence of keywords in the aforementioned 68 papers. The statistical analysis reveals a total of 259 unique keywords, with 67 occurring more than twice, 36 appearing more than three times, and 26 appearing more than four times. Fig. 5 displays these 26 keywords, while an additional 10 keywords, appearing three times each, include *construction workers*, *context*, *cycle time*, *simulation*, *activity analysis*, *construction management*, *automation*, *object recognition*, *hydraulic excavator*, and *image processing*. We analyzed these keywords from three aspects. Firstly, when examining these keywords from an application perspective, the top-ranked ones include tracking, action recognition, identification, pose estimation, and recognition. After conducting an in-depth literature review, we classified the papers into three categories: (1) excavator detection, identification, and tracking, (2) excavator action recognition, and (3) pose estimation. Secondly, the analysis of these keywords from the perspective of usage methods indicates that the most prominent terms are features and deep learning. Based on a comprehensive literature review, we categorized them into three types: marker less-based, marker-based, and deep learning-based. Thirdly, from the standpoint of application purposes, an analysis of these keywords reveals significant terms such as productivity, safety, construction management, and automation. Furthermore, this paper classified these purposes into three aspects: safety management, improving productivity, and automation control. Consequently, the subsequent sections will undertake a qualitative analysis from these specific perspectives.

### 3. Literature review

According to the quantitative analysis in Section 2, a literature review was conducted on the applications, image features, and purposes of CV.

#### 3.1. Computer vision applications

Due to the limited availability of literature on vision-based excavator manipulator pose estimation, particularly when applied to automatic control, this section not only reviews the literature related to manipulator pose estimation but also conducts literature reviews on “Excavator



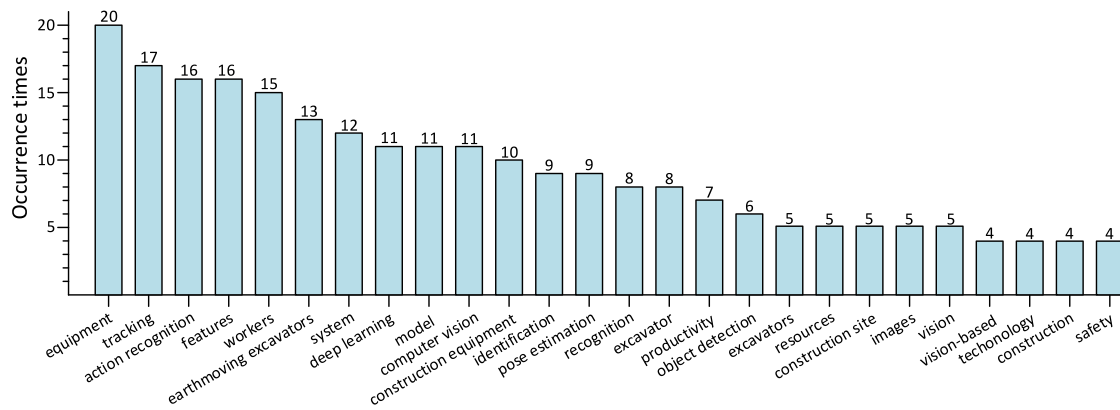


Fig. 5. Occurrence times of keywords

Detection, Identification, and Tracking” and “Action Recognition.” The purpose of this approach is to analyze the diversity of CV methods in the field of excavators and explore the implications for pose estimation.

### 3.1.1. Excavator detection, identification, and tracking

Target identification and target recognition are distinct concepts in the field of CV [37]. In this section, we have not strictly categorized these terminologies due to their shared objective of acquiring excavators in images or videos. Excavator detection refers to the process of identifying and localizing excavators in images or videos by determining their positions using bounding boxes. Excavator tracking aims to establish correspondence between the same excavator in consecutive video frames, enabling the analysis of its dynamic motion. Given the dynamic nature of construction sites, where workers and construction equipment often move irregularly, the detection, identification, and tracking of excavators, as well as other relevant entities such as trucks, workers, and construction materials, are essential for ensuring safety and preventing collisions between excavators and workers. Furthermore, accurate detection of excavators in complex work environments is a prerequisite for subsequently monitoring excavator actions and estimating poses. Early research predominantly focused on sophisticated feature-based detection methods, such as the histogram of oriented gradients (HOG) [38–40], and utilized support vector machines (SVM) for excavator and background classification. Some studies also explored excavator position detection by analyzing the excavator's shape, such as the “V” shape of the arm [21] or other contour features. Additionally, researchers have employed easily recognizable markers like AprilTag, attached to excavators, and utilized the shape features of these markers for excavator detection [41]. With advancements in deep learning frameworks, an increasing number of studies have adopted deep learning techniques. Recent research has employed various versions of convolutional neural networks (CNN), including the region-based fully convolutional network (RFCNN) [17], fast region convolutional neural network (FastRCNN) [42], improved Faster Regions with Convolutional Neural Network Features (IFaster R-CNN) [43], Aff-SSD [9], YOLO [44], among others. However, deep learning-based methods require extensive training datasets, prompting scholars to investigate data collection and annotation methods [45–47]. For more information, refer to Table 2 for a summary of these papers.

### 3.1.2. Excavator action recognition

Excavator action recognition involves analyzing the behaviors and action types of excavators through images, such as digging, dumping, hauling, or swinging. This analysis can enhance operational safety and prevent collisions with trucks and workers. Furthermore, it enables the assessment of normal excavator operation and productivity by utilizing activity data, facilitating effective construction site management. In the early stages, J. Zou [32] utilized CV to analyze the idle time of an

excavator. This approach distinguished the excavator from the background based on color differences and focused on determining whether the excavator was in motion. Similar to CV applications in excavator detection, early applications of action recognition focused on sophisticated feature-based methods such as the Histogram of Oriented Gradients (HOG) and machine learning techniques like Support Vector Machines (SVM) [52]. More recently, deep learning methods have been employed for action recognition, including CNN [18], Simplified Temporal Convolutional Network (STCN) [53], Multistage Temporal Convolutional Network (MS-TCN) [54], and YOLO [20,55]. Table 3 provides a summary of these papers.

### 3.1.3. Excavator manipulator pose estimation

Manipulator pose serves several purposes, including safety accident prevention, tele-operator guidance, and operation trajectory control. Table 4 presents a summary of papers on pose estimation. Before 2014, studies used distinct self-made markers with specific shapes and colors attached to buckets or manipulators for feature recognition, indirectly determining the manipulator pose. For instance, F. Wang [57] used stripes to represent the bucket's position, while a red marker indicated its orientation. H. Wang [22] employed saddle points for acquiring world coordinates. Despite performing well in specific scenarios, the simplicity of markers often led to reduced robustness in complex environments. AprilTag markers were introduced in 2011 and have since been successfully applied in robot navigation, object recognition, and augmented reality. Subsequently, from 2015 to 2018, they gradually gained popularity in excavator pose estimation [58,65,68]. AprilTags are standardized black and white matrix markers, known for their robustness and accurate recognition across various scenarios. Additionally, researchers explored methods based on excavator contours [23,67,69,70]. Due to the markers' relatively small size, issues like occlusion and damage occurred during excavator operations. In contrast, approaches focusing on excavator contour features exhibited higher robustness. Both contour and marker-based methods rely on image features manually specified by humans, as opposed to self-learning. In recent years, CNN-based deep learning models gained traction due to advancements in hardware, larger annotated datasets, and improved algorithms. Deep learning's ability to automatically learn excavator features has become a research hotspot. Consequently, its application in excavator pose estimation, such as [59–63], was published from 2018 to 2022. Notably, the mentioned papers mainly aimed to prevent safety accidents and improve operational efficiency, with only a few utilized as feedback signals for automatic control systems [22,64,66,71].

In the field of preventing safety accidents and improving operational efficiency, deep learning methods hold significant potential. Given sufficient training data, deep learning approaches can effectively learn and adapt to various factors affecting pose estimation, such as excavator operating environments, occlusions, lighting variations, smoke, dust,

**Table 2**  
Literature summary of excavator detection, identification, and tracking.

| Paper | Title                                                                                                                                  | Purpose                                                                                                                                          | Method                                                         | Year |
|-------|----------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|------|
| [38]  | Part based model and spatial-temporal reasoning to recognize hydraulic excavators in construction images and videos                    | Recognizing hydraulic excavators in construction images and videos                                                                               | HOG, SVM , and spatial-temporal reasoning                      | 2012 |
| [39]  | Automated 2D detection of construction equipment and workers from site video streams using histograms of oriented gradients and colors | Automated 2D detection of construction equipment and workers from site video streams to ensure safe operation                                    | HOG + Color+ binary SVM                                        | 2013 |
| [46]  | Image dataset development for measuring construction equipment recognition performance                                                 | Proposed a standard dataset of construction site images for construction equipment recognition                                                   | Developed by Felzenszwalb et al. [48] and Torralba et al. [49] | 2014 |
| [21]  | Hydraulic Excavators Recognition Based on Inverse “V” Feature of Mechanical Arm                                                        | Describing an hydraulic excavator detection system based on inverse “V” feature of mechanical arm for monitoring illegal construction activities | Contour features                                               | 2014 |
| [40]  | Towards Part-Based Construction Equipment Pose Estimation Using Synthetic Images                                                       | Parts are detected to improve production efficiency and safety                                                                                   | HOG                                                            | 2016 |
| [41]  | Construction Equipment Identification Using Marker-Based Recognition and an Active Zoom Camera                                         | Excavator identification using marker-based recognition for improving the productivity                                                           | HOG, Linear SVM, and AprilTag                                  | 2016 |
| [47]  | Automated annotation for visual recognition of construction resources using synthetic images                                           | The recognition of construction equipment for monitoring the progress and the safety                                                             | HOG                                                            | 2016 |
| [17]  | Detecting Construction Equipment Using a Region-Based Fully Convolutional Network and Transfer Learning                                | Excavator detection for producing managerial information such as progress, productivity, and safety                                              | R-FCN                                                          | 2018 |
| [43]  | Automated detection of workers and heavy equipment on construction sites: A convolutional neural network approach, Advanced            | Excavator detection for improve safety and productivity                                                                                          | IFaster R-CNN                                                  | 2018 |

**Table 2 (continued)**

| Paper | Title                                                                                                                        | Purpose                                                                                                                         | Method                                                                                                                                        | Year |
|-------|------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|------|
| [42]  | Engineering Informatics Towards database-free vision-based monitoring on construction sites: A deep active learning approach | Achieved database- free (DB-free) vision-based monitoring on construction sites                                                 | Faster R-CNN,SSD, YOLO                                                                                                                        | 2020 |
| [50]  | Multi-camera vision-based productivity monitoring of earthmoving operations                                                  | Excavator detection to monitor the operational efficiency and productivity                                                      | Tracking-Learning-Detection (TLD)                                                                                                             | 2020 |
| [15]  | Dense construction vehicle detection based on orientation-aware feature fusion convolutional neural network                  | Accurate identification of dense multiple vehicles for ensuring the safety of construction sites                                | Orientation-aware feature fusion single stage detection (OAFF-SSD)                                                                            | 2020 |
| [44]  | Vision-Based Method Integrating Deep Learning Detection for Tracking Multiple Construction Machines                          | Tracking construction machines in videos for automated surveillance of construction productivity, safety, and project progress. | YOLOv3                                                                                                                                        | 2021 |
| [51]  | Temporal image analytics for abnormal construction activity identification                                                   | Excavator identification and tracking for productivity and safety                                                               | Faster Region-proposal Convolutional Neural Network (Faster R-CNN) for detection and Simple Online and Real time Tracking (SORT) for tracking | 2021 |
| [45]  | Development of training image database using web crawling for vision-based site monitoring                                   | Development of training image database using web crawling                                                                       | Faster R-CNN                                                                                                                                  | 2022 |

etc. In contrast, methods based on markers or excavator contour are often heavily influenced by these factors, leading to reduced robustness. Therefore, relevant image processing algorithms are required to address these influencing factors. The literature review in [Section 3](#) indicates that deep learning algorithms show promise in achieving the goal of improving operational safety. Besides challenges related to computer hardware and network models, deep learning faces the task of collecting datasets that encompass various influencing factors. Fortunately, advancements in data sharing and dataset synthesis techniques further facilitate the application of deep learning methods. Overall, deep learning holds potential in addressing the challenges of manipulator pose estimation for safety operation and improving operational efficiency, provided that sufficient and diverse datasets can be obtained.

Considering the challenges and future development directions of these methods in automatic control, we thoroughly discuss these aspects in [Section 4](#).

### 3.2. Computer vision image features

From the perspective of image features, this section categorizes CV methods for excavators into two main aspects: deterministic feature-based methods and deep learning-based methods. The term “deterministic feature-based methods” refers to the use of conventional image

**Table 3**

Literature summary of excavator action recognition.

| Paper | Title                                                                                                                          | Action type                                                             | Purpose                                                                                  | Method                                                      | Year |
|-------|--------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|------------------------------------------------------------------------------------------|-------------------------------------------------------------|------|
| [32]  | Using hue, saturation, and value color space for hydraulic excavator idle time analysis                                        | Moving or not moving                                                    | Improve production efficiency                                                            | Color-based                                                 | 2007 |
| [52]  | Vision-based action recognition of earthmoving equipment using spatio-temporal features and support vector machine classifiers | Digging, dumping, hauling, swing                                        | Improve operational efficiency                                                           | HOG+SVM                                                     | 2013 |
| [18]  | Action recognition of earthmoving excavators based on sequential pattern analysis of visual features and operation cycles      | Digging, hauling dumping, swinging, moving, and stopping                | Productivity analysis                                                                    | CNN and Double layer Long Short Term Memory                 | 2019 |
| [53]  | A Deep Learning-Based Approach to Enable Action Recognition for Construction Equipment                                         | Digging, dumping, swing                                                 | Improve productivity                                                                     | Simplified temporal convolutional network                   | 2020 |
| [55]  | Working stage identification of hydraulic excavator                                                                            | Walking, digging, lifting and swing, unloading, returning and adjusting | Energy saving                                                                            | Parts detection by Yolo and combined with the pump pressure | 2021 |
| [54]  | Synthesizing Pose Sequences from 3D Assets for Vision-Based Activity Analysis                                                  | Idle, swing bucket, Load bucket, dump, move                             | Dataset synthesis                                                                        | Multistage Temporal Convolutional Network (MS-TCN)          | 2021 |
| [20]  | Automatic Identification of Idling Reasons in Excavation Operations Based on Excavator-Truck Relationships                     | Estimated idle time                                                     | Control the work efficiency and productivity                                             | YOLOv3                                                      | 2021 |
| [56]  | Vision-Based Activity Classification of Excavators by Bidirectional LSTM                                                       | Dumping, hauling, excavation, and return                                | Improved the safety, continuous monitoring, productivity, and cost of the earthwork site | CNN-BiLSTM                                                  | 2023 |

processing techniques rather than deep learning methods. Section 4 will provide a more detailed exploration of these topics.

### 3.2.1. Deterministic feature

Deterministic features in CV on excavators refer to predefined image features in the design of the vision scheme. These features can be classified into marker-less-based (self-feature-based) and marker-based methods. Marker-less-based methods use inherent characteristics of excavators, such as edge and texture-related Histogram of Oriented Gradients (HOG) features [38,40,52], color features [32], contour features, or combinations of multiple features [39] extracted from the excavator manipulator [11,21,67,69]. Therefore, they are also referred to as self-features. These features are combined with machine learning algorithms, such as support vector machines, or template matching methods, to create detectors for excavators and their parts. Marker-based methods involve the design of specific markers with distinctive features. These marker features typically include special geometric shapes or contours, such as dedicated AprilTags [41,58,65,68], Aruco markers [74], custom-made saddle point markers [22], specific colors [32,66], or markers that combine multiple features [57,64]. Image processing algorithms, such as corner detection, edge detection, and color threshold, are employed to extract the marker feature points.

### 3.2.2. Deep learning feature

Deep learning-based CV eliminates the design of image features. These algorithms automatically learn target features and establish the mapping relationship between images and the desired quantities. In CV, deep learning algorithms are typically based on CNN, and numerous network models have been developed and refined. Examples include the widely used two-stage object detection algorithm, R-CNN [75], as well as one-stage object detection algorithms like YOLO and SSD [76], along with their modified architectures. Additionally, algorithms such as the Stacked Hourglass Network are utilized for pose estimation. The detailed algorithms are listed in Tables 2 to 4. Deep learning algorithms for excavators eliminate the need for manually designed features by leveraging the network's autonomous ability to learn features. Consequently, a substantial amount of training data is usually required to enhance the network's learning capabilities. Therefore, data annotation and synthesis [45,46] have emerged as important topics in the field of CV for excavators.

Observing the “Year” and “Method” columns in Tables 2 to 4, it is evident that deterministic features were initially employed for excavator

detection and action recognition. However, in recent years, deep learning-based methods have gained widespread popularity. It can be attributed to the maturity of deep learning methods and network models. The availability of open-source datasets, as well as specialized techniques for dataset synthesis and annotation, has enriched the information of training sets. These datasets now encompass excavator models, scales, camera perspectives, poses, backgrounds, lighting conditions, and more. For excavator manipulator pose estimation, although deep learning methods have made progress and found applications in recent years, their representation of the excavator pose as pixel coordinates of junction points [61,62,72,73], similar to human pose estimation, does not typically serve as the required feedback variable for automatic control of excavator operations. As a result, the marker-based method remains popular for pose estimation at present.

### 3.3. Purpose of computer vision on excavators

The applications of CV on excavators can be classified into three main purposes. Firstly, it ensures operational safety. Secondly, it enables equipment management, improves work efficiency, and promotes energy conservation. Lastly, it generates feedback data for automatic control. Excavator detection, identification, and tracking typically involve automatically recognizing excavators in images and working in conjunction with other detectors, such as human and truck detectors, to estimate potential collisions or offer warning signals and avoid hazardous operational behaviors. Excavator detection can also serve as a prerequisite for action recognition. Excavator action recognition provides valuable insights into the types of movements, allowing for monitoring, analysis, and detection of potential issues or errors. It also enables real-time feedback on work status, reliance reduction on manual operators and improvement of operational efficiency and safety. Furthermore, excavator action recognition helps assess the quality of the operations during different work stages. For example, it can evaluate the depth and accuracy of excavation or assess the efficiency of unloading, leading to improved operational techniques and work quality. It is also worth mentioning that recognizing operational movements can serve as a reference for energy-saving technologies in excavators. Different operational movements have varying demands on the power and hydraulic systems, and adjusting them accordingly can optimize energy usage. Excavator pose estimation plays a crucial role in ensuring operational safety, improving work efficiency, and facilitating vision-based automatic control by providing essential data on the excavator

**Table 4**  
Literature summary of excavator manipulator pose estimation.

| Paper | Purpose                                                                                            | Method                                              | Experimental environment                                                    | Results                                                                                                                                                      | Year | Camera information                                                         |
|-------|----------------------------------------------------------------------------------------------------|-----------------------------------------------------|-----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|------|----------------------------------------------------------------------------|
| [57]  | Autonomous mining                                                                                  | Stripe color block and the red mark                 | /                                                                           | /                                                                                                                                                            | 2011 | One color CCD camera placed on the ground facing the bucket                |
| [67]  | For construction monitoring and safety management                                                  | Contour feature and template matching               | Excavator model                                                             | /                                                                                                                                                            | 2014 | One surveillance camera placed outside the excavator                       |
| [22]  | Automation operation                                                                               | Label saddle point (corner point)                   | Real excavator experiment                                                   | The joint Angle error is very small                                                                                                                          | 2014 | One gray industrial camera with an infrared filter placed on excavator cab |
| [58]  | Teleoperation for guiding human operations                                                         | AprilTag                                            | Indoor simulation experiment                                                | Static error is about 0.5°, dynamic error is close to 1°.                                                                                                    | 2015 | One regular digital camera placed on the ground facing the bucket          |
| [68]  | Auxiliary operation                                                                                | AprilTag                                            | Real excavator experiment                                                   | The position error of bucket tip in all three directions is within 3 cm.                                                                                     | 2016 | Three gray optical cameras placed on excavator cab                         |
| [69]  | Enhancing workplace safety in the construction industry                                            | Template matching and contour detection             | Real excavator experiment                                                   | 70% of the errors in the z-axis are within the range of (−1.0 m, 1.0 m). 95% of the errors in the x-axis and y-axis are within the range of (−0.5 m, 0.5 m). | 2017 | Two high-definition camcorders (SONY HDR-CX760V) placed on the cab         |
| [23]  | Improving safety and productivity of construction processes                                        | Part detectors and pose estimation based on contour | Synthetic data set                                                          | The estimated errors of the two joint angles are less than 4° and 8°, with a maximum of 18°.                                                                 | 2017 | Data set                                                                   |
| [70]  | Improving the productivity of operations and safety on job sites                                   | Based on contour.                                   | Real excavator experiment                                                   | 2.9° and 4.2° are the average errors of the estimated angles.                                                                                                | 2018 | Four color surveillance cameras                                            |
| [65]  | Improving both safety and productivity in many construction tasks                                  | AprilTag                                            | Real excavator experiment                                                   | The accuracy of the vertical orientation of the bucket tooth tip is 2.54 cm                                                                                  | 2018 | Two color industrial cameras placed on the cab                             |
| [71]  | Automation operation                                                                               | Neural network                                      | Simulation experiment                                                       | Error for the boom cylinder displacements from −0.263 cm to 0.281 cm, arm from −1.230 cm to 0.780 cm, and bucket from −1.878 cm to 1.387 cm                  | 2019 | One color webcam placed outside the excavator facing the excavator         |
| [59]  | For safety on construction sites                                                                   | Stacked hourglass network                           | Real manipulator experiment                                                 | The mean Euler distance error of the Boom-Stick node is 63.81 mm. Stick-Bucket is 37.01 mm. Bucket is 52.16 mm                                               | 2019 | Outside excavator (Data set)                                               |
| [60]  | For safety on construction sites                                                                   | Stacked hourglass network                           | Real manipulator experiment                                                 | Precision 144.65 mm                                                                                                                                          | 2019 | Outside excavator (Data set)                                               |
| [64]  | Automation operation                                                                               | Color marker                                        | Real excavator experiment                                                   | The world coordinate error of the static marker within 30 mm                                                                                                 | 2020 | One color industrial camera placed on the ground facing the excavator      |
| [61]  | Real-time safety monitoring on construction sites                                                  | Stacked hourglass network, Cascaded pyramid network | Image dataset                                                               | HG and CPN can achieve a PCK (percentage of correct key-points) value of 91.19% and 91.78%.                                                                  | 2020 | Outside excavator (Data set)                                               |
| [72]  | Creating a safer and more productive work environment.                                             | HRnet                                               | Image dataset                                                               | Synthetic data set NE = 0.0498. Real data set NE = 0.0478                                                                                                    | 2022 | Outside excavator (Data set)                                               |
| [62]  | Safety monitoring                                                                                  | YOLOv5-FastPose (YFP)                               | Alberta construction image dataset and construction equipment poses dataset | Average normalization error (NE) of 0.01294                                                                                                                  | 2022 | Outside excavator (Data set)                                               |
| [63]  | For increasing awareness about jobsite hazards and operational performance in earthwork operations | CNN                                                 | Real excavator experiment                                                   | The average pose error is 9.63 degrees                                                                                                                       | 2021 | One smartphone placed on the ground                                        |
| [73]  | For edge device deployment                                                                         | Lightweight fully convolutional network (FCN)       | Real excavator experiment video                                             | The maximum average precision is 0.927                                                                                                                       | 2022 | One surveillance camera                                                    |
| [66]  | Automation operation                                                                               | Color Marker                                        | Real excavator experiment                                                   | The maximum joint angle error is −1 to +1 degree                                                                                                             | 2022 | One Color camera fixed on the cab                                          |
| [74]  | Improve operational safety                                                                         | Aruco marker                                        | Real excavator experiment                                                   | Average RMSEs 119.85 mm                                                                                                                                      | 2023 | Two smartphones placed in the cab                                          |

manipulator's position relative to its base.

## 4. In-depth discussion

### 4.1. Content of in-depth discussion

Considering that most papers on vision-based excavator detection and action recognition primarily focus on the application of new technologies, such as deep learning or image processing, it is important to

acknowledge that excavators are not merely objects of study. The measurement of excavator poses using vision-based methods is influenced not only by these technologies but also by the structures and operational characteristics of excavators. Consequently, vision-based pose measurement methods encounter numerous challenges and difficulties when applied to automatic excavator operations. Additionally, the literature review in [Section 3.1.3](#) highlights that the pose measurements are used for ensuring safe operation and productivity assessment, as well as for automatic control. Unlike the former, the latter imposes



stringent requirements on measurement accuracy and speed. Therefore, the objective of this section is to provide a comprehensive discussion of vision-based pose measurement methods for excavator manipulators, with a specific emphasis on the perspective of automatically controlling operation trajectories.

Firstly, the definition of vision-based manipulator pose estimation for heavy-duty hydraulic excavators is introduced, highlighting its distinctions from conventional industrial robots. The specific differences are as follows. (1) Camera positioning: Due to the working specialties of heavy-duty hydraulic excavators, cameras are typically installed outside the manipulators (eye-to-hand). Common mounting locations include fixed brackets on the ground [59], building walls [70], or on the excavator cabs [68] to ensure the cameras remain undamaged. The excavator cabs are optimal positions since they can move or rotate along with the walking and swing mechanisms, as demonstrated in Fig. 6. In contrast, conventional industrial robots usually operate in static working sites where cameras are fixed at the end of manipulators (eye-in-hand) or in locations with consistent pose relationships relative to their base. (2) Applications of pose estimation. As illustrated in Fig. 6,  $\{E\}$ ,  $\{B\}$ ,  $\{C\}$ , and  $\{O\}$  represent the coordinate systems of the manipulator end, base, camera, and object, respectively. The coordinate transformations between  $\{E\}$  and  $\{B\}$  are denoted as ETB, and between  $\{E\}$  and  $\{O\}$  as ETO. In the field of excavators, our focus is on the vision-based measurement of ETB, which can be represented by the bucket end coordinates and bucket orientation relative to the base, the displacements of the three actuator cylinders, and the relative joint angles between the links. There is a unique conversion relationship between these representations. While contact-sensor-based measurement methods for ETB involve contact sensors, Section 1 mentions the potential of vision-based measurements as a novel approach. For industrial robots, pose estimation typically involves estimating the pose between the camera and the object (CTO). For eye-to-hand configurations, CTB can be obtained through hand-eye calibration, while ETB is primarily measured using sensors such as motor encoders and angle sensors. Finally, ETO is commonly achieved, allowing the end effector of the industrial robot to approach and grasp the object. (3) Degrees of freedom of manipulators. In excavators equipped with buckets, vibratory compactors, breaking shears, and similar attachments, the manipulators typically operate within a single plane, which means the rotation axes of the manipulators are parallel. In contrast, industrial robots usually have non-parallel rotation axes. This unique mechanical structure makes hand-eye calibration methods used for industrial robots unsuitable for excavators [66]. Furthermore, there are additional differences in terms of working environments, tasks, and application purposes. Consequently, this paper will not delve into detailed discussions of pose estimation technology related to industrial robots.

Next, the concept of feedback control is introduced. Feedback control involves providing the computer with real-time manipulator pose as feedback data (ETB), which is used to calculate the deviation from the target signal. Based on this feedback, control algorithms are employed to correct the deviation and track the desired quantity. Therefore, at this stage, the manipulator pose is obtained through visual measurement methods, rather than relying on commonly used contact sensors. In this paper, the term “MPVF” is adopted to represent the Manipulator Pose of a Vision-based method for Feedback control.

After providing a clear explanation of the relevant concepts, this section will undertake a comprehensive discussion, addressing the following aspects: analyzing the requirements for MPVF measurement, evaluating the existing limitations of CV methods in relation to MPVF, and proposing future trends for the implementation of MPVF measurement.

#### 4.2. Requirement analysis and literature evaluation

The vision-based manipulator pose plays a crucial role in the automatic control of the operation trajectory, replacing the function of contact sensors. Consequently, vision-based measurement methods are required to meet several requirements. Firstly, rapidity. The data update frequency should be high enough to provide real-time poses. Secondly, accuracy. The vision-based pose estimation should exhibit minimal error compared to traditional contact sensors. Thirdly, reliability. The poses should remain consistent and not diverge during control. Lastly, less cost. Economic factors, calibration, maintenance, and time or labor costs must be considered, ensuring the method's feasibility. This section provides a deep analysis and discussion of these four aspects. Furthermore, the literature mentioned in Section 3.1.3 is evaluated and summarized in Table 5. Due to the limited availability of literature specifically focused on trajectory automatic control, all the papers mentioned in Section 3.1.3 are evaluated to analyze the diversity of methods. It is important to note that the four requirements mentioned earlier are proposed in the field of automatic control. However, the papers mentioned in Section 3.1.3 primarily focus on safety monitoring and productivity detection, rather than trajectory automatic control. Therefore, the evaluation of these papers in Table 5 from the perspective of automatic control does not negate or diminish the research findings presented in those studies.

##### 4.2.1. Rapidity

Automatic control places a strong emphasis on the real-time performance of measurements [77]. This implies that the measurement frequency should be rapid and without significant delays. If the real-time performance is inadequate, the current pose information fed back

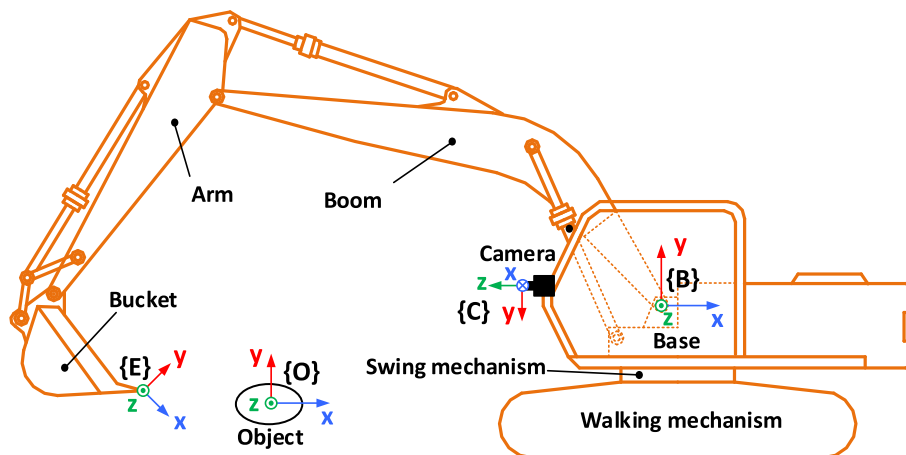


Fig. 6. Structure of heavy-duty hydraulic excavator.

**Table 5**

Qualitative evaluation of visual-based excavator pose measurement papers for automatic control application (o and x represent strong and weak respectively).

| Paper | Title                                                                                                                                            | Method                                              | Year | Rapidity | Accuracy | Reliability | Time-cost |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|------|----------|----------|-------------|-----------|
| [57]  | The Recognition of Target and Attitude of Bucket for Excavator Robot Based on Color Mark Tracking                                                | Stripe color block and the red mark                 | 2011 | o        | o        | x           | o         |
| [67]  | Key Nodes Modeling for Object Detection and Location on Construction site using Color-Depth Cameras                                              | Contour feature and template matching               | 2014 | x        | x        | o           | x         |
| [22]  | A Novel Approach to Detect the Posture of Excavator's Manipulator                                                                                | Label saddle point (corner point)                   | 2014 | o        | o        | x           | o         |
| [58]  | Feasibility of In-plane Articulation Monitoring of Excavator Arm Using Planar Marker Tracking                                                    | AprilTag                                            | 2015 | o        | o        | x           | o         |
| [68]  | Optical Marker-based End Effector Pose Estimation for Articulated Excavators                                                                     | AprilTag                                            | 2016 | o        | o        | x           | x         |
| [69]  | Vision-Based Excavator Detection and Tracking Using Hybrid Kinematic Shapes and Key Nodes                                                        | Template matching and contour detection             | 2017 | x        | x        | o           | x         |
| [11]  | Skeleton Estimation of Excavator by Detecting Its Parts                                                                                          | Based on skeleton                                   | 2017 | x        | x        | x           | o         |
| [70]  | Framework for Location Data Fusion and Pose Estimation of Excavators Using Stereo Vision                                                         | Based on contour                                    | 2018 | x        | x        | x           | o         |
| [65]  | Camera Marker Networks for Articulated Machine Pose Estimation                                                                                   | AprilTag                                            | 2018 | o        | o        | x           | x         |
| [71]  | Vision-based Estimation of Excavator Manipulator Pose for Automated Grading Control                                                              | Neural network                                      | 2019 | o        | o        | x           | x         |
| [59]  | Fast Dataset Collection Approach for Articulated Equipment Pose Estimation                                                                       | Stacked hourglass network                           | 2019 | o        | x        | o           | x         |
| [60]  | A Vision-based Marker-less Pose Estimation System for Articulated Construction Robots                                                            | Stacked hourglass network                           | 2019 | o        | x        | o           | x         |
| [64]  | The Position Measurement of Excavator Working Device                                                                                             | Color marker                                        | 2020 | o        | x        | x           | o         |
| [61]  | Full Body Pose Estimation of Construction Equipment Using Computer Vision and Deep Learning Techniques                                           | Stacked hourglass network, Cascaded pyramid network | 2020 | x        | x        | o           | x         |
| [63]  | Implementation Experiments on Convolutional Neural Network Training Using Synthetic Images for 3D Pose Estimation of an Excavator on Real Images | CNN                                                 | 2021 | x        | x        | o           | x         |
| [72]  | Vision-based Excavator Pose Estimation Using Synthetically Generated Datasets with Domain Randomization                                          | HRnet                                               | 2022 | x        | x        | o           | x         |
| [62]  | Pose Estimation Method for Construction Machine Based on Improved AlphaPose Model                                                                | YOLOv5-FastPose(YFP)                                | 2022 | x        | o        | o           | x         |
| [73]  | Excavator Joint Node-based Pose Estimation Using Lightweight Fully Convolutional Network                                                         | Lightweight fully convolutional network (FCN)       | 2022 | o        | o        | o           | x         |
| [66]  | A New Measurement Method of Real-time Pose Estimation for an Automatic Hydraulic Excavator                                                       | Color Marker                                        | 2022 | o        | o        | x           | o         |
| [74]  | Full-body Pose Estimation for Excavators Based on Data Fusion of Multiple Onboard Sensors                                                        | Aruco marker                                        | 2023 | x        | x        | o           | o         |

to the controller may not reflect the true current status. For instance, M. Soltani [11] took approximately 3–4 s to process a frame image completely, while contact sensors typically operate at frequencies ranging from several hundred to several thousand hertz. Similarly, [70] exhibited an image processing speed of about 2 s per frame, and [65] demonstrated varying processing speeds depending on the image size, with 640\*480 resolution at 20 Hz and 1280\*1960 resolution at 5 Hz. Significant changes in the manipulator pose may occur during these delays. Consequently, achieving accurate manipulation becomes challenging when controlling the current pose with the delayed data. The low measurement frequency is influenced by both the camera's sampling speed and the computational speed of the image processing unit, and it can also be influenced by the measurement algorithms. Due to the limited computational performance of hardware systems in practical applications, image processing algorithms should also have minimal time consumption. Based on the literature review in Section 3.2, current vision-based pose measurement methods on excavators can be categorized as marker-based, self-feature-based, and deep-learning-based. Therefore, the factors influencing rapidity will be discussed from these perspectives, as explored in the following sections.

- (1) The measurement delay based on deep learning primarily arises from the complexity of the algorithm framework. A construction site image, as depicted in [73], consists of a large number of pixels, such as 384\*2546\*3 pixels. Moreover, deep learning frameworks typically involve convolution, pooling, and fully connected layers applied to each frame image [78]. Deep learning models often comprise numerous layers of convolution and

pooling, ranging from dozens to hundreds, or even more, which significantly increases the data processing load for each image.

- (2) For marker-based and self-feature-based measurement methods, the complexity of image features can necessitate complex image processing, which in turn leads to a time-consuming process per frame. As a consequence, the frequency of pose estimation is reduced. In summary, under the same hardware situation, compared with deep-learning-based methods, marker-based and self-feature-based methods may offer greater ease in ensuring rapidity due to their inherent characteristics.

#### 4.2.2. Accuracy

To achieve high-precision trajectory control, it is crucial to ensure the data accuracy provided to the controller. As contact sensor measurement technology is well-established, the accuracy of vision-based measurements can be evaluated by comparing them with the data obtained from contact sensors.

In the case of deep learning-based measurement methods, it is important to ensure good generalization performance of the detector. This means that the trained deep learning network should not only perform well on the training dataset but also accurately estimate poses for new datasets. The accuracy of the method is influenced by several factors, including the structural framework of deep learning, training parameters, diversity of training sets, and the generalization ability of the trained network. For example, in [63], the CNN method showed an average prediction error of 9.63° for joint angles. While this level of accuracy may be sufficient for detecting movements of excavators and ensuring safety, it falls short for precise automatic control. One of the

main reasons for this discrepancy is that the training set in [63] is artificially generated, rather than derived from real on-site data. Furthermore, for automatic control purposes, it is typically necessary to obtain a specific form of pose data, such as joint angles and cylinder displacement. However, most deep learning methods [59,60,62] only provide coordinates of the hinge point in the image, which cannot be directly utilized for control. Additionally, the evaluation of these methods is limited to discrete images from the training set, lacking real-time validation during actual excavator operations.

In marker-based and self-feature-based methods, inappropriate image processing may result in inadequate adaptation of marker feature identification to variations in lighting conditions, dust interference, short-term obstructions, and other issues. For example, although the data presented in papers [22,58,64,66,68] demonstrate precision, they did not consider the impact of environmental changes. Furthermore, the accuracy of the mapping relationship between the image and pose information [79] is a crucial factor affecting the overall accuracy. This mapping model typically comprises a homogeneous coordinate transformation model [80] and a perspective projection model [81]. However, there are several factors that can introduce errors in the mapping model. (1) The mapping model is an approximation that disregards or approximates numerous details, such as lens distortion [82] or the keyhole imaging model. These simplifications can introduce modeling errors. (2) Errors can occur in the calibration of model parameters. Traditional calibration methods, such as the Zhang method [83], typically provide relatively accurate calibration parameters. However, errors may arise when the calibration plate is far away from the operating space, particularly for large-sized excavators. Offline calibration may not fully cover the entire range of the bucket operating space, making it difficult to adapt the obtained parameters to all operating trajectories. (3) Pose estimation for excavators usually involves estimating the world coordinates of feature points and performing indirect calculations, as observed in various literature sources [22,64,66]. However, the estimation of world coordinates is highly sensitive to model parameters. Consequently, errors in these pose measurement methods propagate to the estimated pose, posing challenges in meeting accuracy requirements. Moreover, marker features in images may appear blurred due to the relatively fast operation speed of the manipulator. These influencing factors need to be considered to enhance the overall accuracy of pose measurement methods.

#### 4.2.3. Reliability

Reliability refers to the stability of measurements, ensuring that there is no long-term loss of measurement data. Continuous and uninterrupted feedback data is crucial because if the controller fails to acquire the latest pose, the operation trajectory cannot be accurately controlled. This can result in incorrect operation and pose a safety risk to workers. Several factors can contribute to the unreliability of pose estimation. In some deep learning-based methods, the camera's field of view may not fully cover the entire range of the excavator's operation trajectory, causing certain manipulator poses to fall outside the camera's field of view. Additionally, in marker-based and self-feature-based methods, occlusion of marker features, particularly due to corruption or damage, can introduce unreliability. For instance, H. Ma [64] attached a marker to the bucket's end, while H. Wang [22] affixed saddle points on the bucket linkage. Although these markers have minimal impact during level operations, they may become blocked by soil or sustain damage during digging, rendering vision-based pose measurement methods unusable in such situations. Furthermore, although fixing the camera on the cab [22,64] ensures method usability even when the excavator's operating location changes or the rotating mechanism moves, it is important to consider the limited perspective of ordinary industrial cameras. If the manipulator and markers exceed the camera's field of view, the method may fail, restricting the spatial working range of the actuator. To enhance reliability, it is crucial to ensure complete coverage of the manipulator's trajectory within the camera's field of

view and address potential occlusion or corruption issues associated with marker-based and self-feature-based methods.

#### 4.2.4. Less time investment

In addition to considering economic costs, the vision-based measurement approach should also prioritize time efficiency. Excavators consist of a swing mechanism and a walking mechanism, as shown in Fig. 6. Unlike conventional industrial robots, excavators operate in various locations, which poses challenges for some existing measurement methods. These methods position cameras or markers on the ground [57,58,64,65,68,69]. For instance, [57,58] require the camera to be positioned as perpendicular as possible to the excavator's working plane or within a certain angle range. Consequently, when the excavator changes its working site or undergoes rotation, the feature points may fall outside the camera's field of view. Under such circumstances, these methods become inapplicable unless the marker and camera network undergo recalibration. However, recalibration can be even more time-consuming than the manipulation tasks themselves. Moreover, pose estimation methods based on deep learning typically rely on large datasets [46] to cover various conditions, such as excavator postures, lighting conditions, and background dust and smoke, in order to achieve high accuracy. The estimation of manipulator pose is influenced by factors such as the excavator model, camera model, and camera installation position. Therefore, it is essential to collect images that encompass diverse environments and manipulator postures for each type of camera, camera parameter, installation position, and excavator model. This process involves creating an extensive training dataset that contains a significantly large number of image samples. Additionally, manual annotation is sometimes necessary, making the data collection and processing procedures time-consuming and laborious. Furthermore, the training process of deep learning models, in general, is complex and time-consuming.

#### 4.3. Future development

This section discusses future developments in the pose measurement of excavators, focusing on marker-based methods, deep learning-based methods, and the four requirements proposed in Section 4.2.

##### 4.3.1. Marker-based method

- (1) Feature point occlusion problem: Given the focus of this paper on vision-based measurement methods, non-occlusion is a fundamental requirement. Therefore, the measurement schemes should aim to minimize the influence of marker occlusion by considering factors such as camera installation positions, camera angles of view, and marker placement. Furthermore, it is necessary to develop fault-tolerant algorithms capable of handling transient occlusion of image features. These algorithms should accurately, continuously, and completely track feature points, ensuring their preservation. Predictive algorithms like the Kalman filter [84] and particle filter [85] can be employed when a marker is temporarily blocked to estimate the manipulator pose. If markers need to be placed on the bucket, it will inevitably affect the operation. Once the excavator digs the ground, the bucket can become buried, leading to potential damage or obstruction of the markers by mud. This situation is not conducive to subsequent work. In such cases, special marker materials that prevent mud adhesion might be a good method. While for hydraulic rippers and breaking hammers, their upper parts are typically positioned above the ground during operation, reducing the likelihood of marker burying issues. However, hydraulic shears and timber grabs, which can rotate similarly to wrists, may encounter difficulties with marker visibility in the camera view. To ensure marker visibility during operation, the use of multiple markers on these specific excavator attachments may be crucial. Moreover,

to enhance the spatial operating range of the actuator as observed by the camera, one potential approach is to install multiple cameras in the cab with different perspectives. This approach aims to comprehensively cover the entire range of the actuator's movements. However, this approach presents challenges in terms of pose calibration among multiple cameras and matching the same marker feature point from various camera perspectives. Alternatively, considering the use of a wide-angle camera can address the aforementioned issues. Nevertheless, establishing an accurate distortion model and calibrating precise distortion parameters may cause significant challenges.

- (2) Marker features: The features of markers play a crucial role in distinguishing objects from the background. Geometric features, color features, and texture features can be utilized as the basis for marker recognition, with a focus on developing markers with enhanced recognition capabilities. By fusing multiple features, the robustness of feature point recognition can be improved, particularly in challenging environmental conditions such as by lighting variations, smoke, and dust particles.
- (3) Mapping relationship: As discussed in Section 4.2.2, the impact of the mapping model on pose measurements primarily stems from model errors and parameter calibration errors. Therefore, in future research, offline calibration techniques that are independent of modeling, such as neural networks [65], could be considered. These techniques employ nonlinear fitting to adapt to inaccurate models and parameter variations resulting from changes in the operating location. Furthermore, it is worthwhile to explore more precise modeling techniques for mapping models, such as small hole imaging models and distortion models. In such cases, achieving precise optimization of model parameters becomes crucial, although the complexity of the model also presents challenges for nonlinear optimization algorithms. Intelligent optimization algorithms may offer a viable solution to address these challenges.

#### 4.3.2. Deep learning-based method

Deep learning-based methods hold potential for monitoring purposes to prevent hazards, manage workers, and enhance production efficiency. In the field of manipulator pose measurement for feedback control, methods based on deep learning demonstrate significant potential. For instance, deep learning network architectures founded on CNN and Vision Transformers [86] relying on a global attention mechanism showcase substantial promise. The progress in computational hardware further facilitates the real-time application of deep learning in measurement tasks. Deep learning methods eliminate the need for humans to specify image features such as color, geometric shapes, and textures. Instead, these methods rely on the learning abilities of deep networks to automatically learn image features. The discussion below explores the future prospects of vision-based measurement methods using deep learning for excavator manipulator pose measurement from three different aspects.

From a methodological perspective: (1) The one-step measurement method, which employs deep learning techniques to learn the mapping between images and corresponding manipulator poses (joint angles), eliminates the need for establishing homogeneous transformation models and calibrating the camera's intrinsic and extrinsic parameters. However, the one-step method incorporates information about the camera's pose (extrinsic parameters) within the deep neural network. Consequently, if there is a change in the camera's pose, the original dataset becomes unusable. (2) The two-step measurement method, which involves first learning key points from images and then mapping these key points to pose angles. For instance, learning pixel coordinates of key points from images and then mapping these pixel coordinates to joint angles using homogeneous transformation models. Current research [61] primarily focuses on predicting pixel coordinates of key points, while mapping these coordinates accurately to joint angles

requires the establishment of precise mapping models as well as accurate calibration of camera intrinsic and extrinsic parameters. This method is sensitive to errors when predicting a limited number of key points. Therefore, in cases where the excavator CAD model is available, rendering the CAD model to obtain synthetic datasets becomes a viable option. For each rendered image, we can annotate the 3D coordinates corresponding to individual pixels. Subsequently, after deep learning, the 3D coordinates of each pixel can be used to calculate joint angles. This approach can enhance measurement accuracy through multi-point pose estimation.

From the perspective of deep network architecture: It is possible to introduce a detector based on the Swin Transformer [87], which can better fuse information between various features through a global attention mechanism. Additionally, refinement techniques can be borrowed, such as using Recurrent Neural Networks (RNN) or some manually designed detection metrics for further optimization or validation of prediction results. Furthermore, methods from temporal networks can be employed to enhance the accuracy of estimation by fusing and filtering information from multiple frames of video sequences.

From other perspectives: (1) For regression problems of excavator manipulator pose estimation, the use of deep learning methods relies on ensuring the quality and quantity of the dataset. Excavator operations take place in diverse and often challenging outdoor conditions. Collecting training sets comprising images and labels may be a challenging task. In the future, data augmentation techniques can be employed to simulate datasets encompassing complex backgrounds, varying lighting conditions, occlusions, dust, smoke, and more, thus emulating various working environments. However, it is crucial to ensure the authenticity and diversity balance of the augmented data. (2) Because deep learning models often require significant computational power due to their depth and complexity, hardware acceleration techniques like Graphics Processing Unit (GPU) or Tensor Processing Unit (TPU), as well as distributed computing resources or cloud computing platforms, can be leveraged to expedite model training.

## 5. Conclusions

This paper presents a comprehensive review of applications, challenges, and developments in vision-based excavator pose estimation for automatic control purposes. Initially, 174 papers were acquired from the Web of Science database, and after a secondary screening process, 68 papers were selected for further analysis. Quantitative analysis was then conducted on the publication years, journal sources, and countries of these selected papers. The findings revealed *Automation in Construction* and *Journal of Computing in Civil Engineering* as leading journals in the field of computer vision applications for excavators. Based on keyword analysis and content evaluation, CV applications for excavators were categorized into three main aspects: (1) excavator detection, identification, and tracking, (2) excavator action recognition, and (3) excavator pose estimation. Each of these categories underwent a detailed literature review to discuss the trends in these methods. Furthermore, this work provides a literature review that identifies the requirements and challenges of vision-based measurement methods for automatic control of operation trajectory. These requirements include rapidity, accuracy, reliability, and less time investment. The literature on vision-based pose measurement discussed in Section 3.1.3 was evaluated based on these requirements. Marker-based and deep learning-based methods were analyzed, highlighting their advantages and disadvantages. Marker-based methods offer simplicity as they do not require extensive dataset preparation, and the image processing algorithms involved are straightforward. However, these methods have certain limitations. Markers are prone to occlusion, and in specific poses, especially when the camera is far from the marker, the features may not be clearly visible. The development of marker-based methods also faces challenges in the conversion model and parameter solving for transforming image information into manipulator pose. Deep learning-based methods rely



on large datasets, and inadequate samples can lead to inaccurate results. The CNN network structures employed in these methods tend to be complex, and computation speed is often slower. Nevertheless, the advantage of deep learning-based methods lies in their ability to eliminate the need for modeling, feature extraction, and conversion steps. By training a model, the relationship between images and manipulator poses can be mapped. Finally, the paper concludes by discussing future developments in marker-based and deep learning-based visual pose measurement. This paper aims to propose research ideas for the development of vision-based manipulator pose measurement methods, thereby enabling vision-based automatic operations.

For marker-based methods, future research directions include: (1) Developing multi-feature markers based on color, texture, geometric shapes, etc., and fusing multiple features to enhance the robustness of feature point recognition, especially in challenging environmental conditions such as lighting variations, smoke, and dust particles. (2) Accurate modeling of the mapping between feature points and manipulator poses, as well as precise calibration of model parameters, are critical factors for successful application of marker-based methods in automatic control systems. Nonlinear fitting techniques, represented by neural network technologies, may adapt to inaccuracies in modeling and changes in model parameters caused by variations in operational positions. (3) Developing feature point fault-tolerant algorithms to ensure the effectiveness of the measurement algorithm during transient occlusions of markers. (4) To expand the working range of the visual system, the use of camera networks consisting of multiple cameras or wide-angle cameras can be explored to increase the camera field of view, enabling excavators to operate in all directions.

Regarding deep learning-based methods, future research directions include: (1) The one-step measurement method, which employs deep learning techniques to learn the mapping between images and corresponding manipulator poses (joint angles). (2) The two-step measurement method, which involves first learning key points from images and then mapping these key points to pose angles; here, mapping these coordinates accurately to joint angles requires the establishment of precise mapping models as well as accurate calibration of camera intrinsic and extrinsic parameters. (3) It is possible to introduce a detector based on the Swin Transformer, which can better fuse information between various features through a global attention mechanism. (4) Refinement techniques can be borrowed, such as using RNN or some manually designed detection metrics for further optimization or validation of prediction results. (5) Methods from temporal networks can be employed to enhance the accuracy of estimation by fusing and filtering information from multiple frames of video sequences. (6) For regression problems of excavator manipulator pose estimation, the use of deep learning methods relies on ensuring the quality and quantity of the dataset. Data augmentation techniques can be employed to simulate datasets encompassing complex backgrounds, varying lighting conditions, occlusions, dust, smoke, and more. (7) Hardware acceleration techniques like GPU or TPU, as well as distributed computing resources or cloud computing platforms, can be leveraged to expedite model training.

### Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

### Data availability

No data was used for the research described in the article.

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